

# Computationally Efficient Markov Chain Simulation:

Efficient Gibbs Samplers, Metropolis Jumping Rules, & Hamiltonian Monte Carlo

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## Introduction

We want to understand how to make samplers like Gibbs, Metropolis and Metropolis–Hastings *efficient*, i.e. how to reduce autocorrelation so that fewer iterations are needed.

Efficiency is measured by the effective sample size

$$n_{\text{eff}} = \frac{mn}{1 + 2 \sum_{t=1}^{\infty} \rho_t},$$

where  $\rho_t$  is the lag- $t$  autocorrelation,  $m$  the number of chains and  $n$  the draws per chain. Slow mixing arises from strong posterior dependence between parameters, poor scaling, and funnel-shaped geometry (typical in hierarchical models).

## 1 Efficient Gibbs samplers

Gibbs updates each block from its full conditional. It mixes poorly when components are highly correlated, since each conditional move is small relative to the posterior spread.

### 1.1 Transformations and reparameterization

Apply an invertible (often linear) transformation  $\phi = A\theta$  so that the components of  $\phi$  are nearly uncorrelated under the posterior. If  $\Sigma$  approximates the posterior covariance, choosing  $A = \Sigma^{-1/2}$  yields near-independent coordinates, and Gibbs on  $\phi$  mixes quickly. In regression, this corresponds to centering and orthogonalizing predictors.

### 1.2 Auxiliary variables

Introduce latent variables  $\omega$  so that the joint  $p(\theta, \omega \mid y)$  has simple conditionals, while the  $\theta$ -marginal is the target.

**Example 1.1** (Student- $t$  regression). The model  $y_i \sim t_\nu(X_i\beta, \sigma^2)$  is a scale mixture of normals:

$$y_i \mid V_i \sim N(X_i\beta, V_i), \quad V_i \sim \text{Inv-}\chi^2(\nu, \sigma^2).$$

Conditional on  $V$ , the model is weighted normal regression, so  $\beta$  and  $\sigma^2$  have closed-form conditionals, and each  $V_i$  is also inverse- $\chi^2$ .

### 1.3 Parameter expansion

Add a redundant parameter that is not identified by the likelihood but improves mixing. For the hierarchical normal model

$$y_{ij} \sim N(\mu + \theta_j, \sigma_y^2), \quad \theta_j \sim N(0, \sigma_\theta^2),$$

write  $\theta_j = \xi \eta_j$  with  $\eta_j \sim N(0, \sigma_\eta^2)$ , so that  $\sigma_\theta = |\xi| \sigma_\eta$ . The sampler updates  $(\xi, \eta, \sigma_\eta)$  in the expanded space and recovers  $(\theta, \sigma_\theta)$  by deterministic transformation. The extra parameter lets the chain move along the problematic ridge where  $\sigma_\theta$  is near zero and the  $\theta_j$  are near zero.

*Remark.* Conceptually, auxiliary-variable and parameter-expansion methods are related to data augmentation and slice sampling [2].

## 2 Efficient Metropolis jumping rules

Recall the Metropolis step. Given the current  $\theta^{t-1}$ , propose  $\theta^* \sim J_t(\cdot \mid \theta^{t-1})$  and accept with probability

$$\alpha = \min\left(1, \frac{p(\theta^* \mid y) J_t(\theta^{t-1} \mid \theta^*)}{p(\theta^{t-1} \mid y) J_t(\theta^* \mid \theta^{t-1})}\right),$$

which reduces to  $\min(1, p(\theta^* \mid y)/p(\theta^{t-1} \mid y))$  for symmetric  $J_t$ .

### 2.1 Random-walk proposals

Take  $J(\theta^* \mid \theta) = N(\theta^* \mid \theta, c^2 \Sigma)$  with  $\Sigma$  an estimate of the posterior covariance. For a  $d$ -dimensional approximately normal target, the efficient scale is

$$c \approx \frac{2.4}{\sqrt{d}},$$

giving an acceptance rate of about 0.44 for  $d = 1$ , falling to about 0.23 as  $d$  grows. Under these settings, the efficiency of the random walk relative to independent draws is roughly 0.3 in high dimension, and it is lower if  $\Sigma$  is poorly chosen.

### 2.2 Practical tuning

**Practical tuning.**

- Estimate  $\Sigma$  from an initial run (or from a modal approximation), then rerun with the tuned proposal.
- Adaptation should occur only during warm-up; draws used for inference must come from a fixed (non-adapting) kernel, otherwise the chain is no longer Markov with the correct invariant distribution.
- Too small  $c$ : high acceptance but slow diffusion. Too large  $c$ : frequent rejection and stuck chains.

### 3 Further extensions to Gibbs and Metropolis

Some interesting applications which build on the Gibbs and Metropolis–Hastings framework (each is an active area of research and would be discussed in later notes) include:

- **Slice sampling**: uniform sampling under the graph of the density via an auxiliary height variable; adapts step size automatically.
- **Reversible jump MCMC**: for variable-dimension models.
- **Simulated and parallel tempering**: run chains on flattened versions  $p(\theta|y)^{1/T}$  to cross between separated modes.
- **Particle filtering**: sequential Monte Carlo for dynamic models.

### 4 Hamiltonian Monte Carlo

HMC avoids random-walk behaviour by using gradient information to propose distant, high-acceptance moves.

#### 4.1 Setup

Augment  $\theta \in \mathbb{R}^d$  with a momentum  $\phi \in \mathbb{R}^d$  and target the joint

$$p(\theta, \phi | y) = p(\theta | y) p(\phi), \quad \phi \sim \mathcal{N}(0, M).$$

Define the Hamiltonian (“potential” plus “kinetic” energy)

$$H(\theta, \phi) = -\log p(\theta | y) + \frac{1}{2} \phi^\top M^{-1} \phi.$$

Hamilton’s equations are

$$\frac{d\theta}{d\tau} = \frac{\partial H}{\partial \phi} = M^{-1} \phi, \quad \frac{d\phi}{d\tau} = -\frac{\partial H}{\partial \theta} = \frac{d}{d\theta} \log p(\theta | y).$$

These dynamics preserve  $H$ , volume, and are reversible, which is what makes the Metropolis correction valid.

#### 4.2 Leapfrog integrator

Exact trajectories are not available, so discretize with step size  $\epsilon$ . One leapfrog step:

$$\begin{aligned} \phi &\leftarrow \phi + \frac{1}{2} \epsilon \frac{d \log p(\theta | y)}{d\theta}, \\ \theta &\leftarrow \theta + \epsilon M^{-1} \phi, \\ \phi &\leftarrow \phi + \frac{1}{2} \epsilon \frac{d \log p(\theta | y)}{d\theta}. \end{aligned}$$

### 4.3 One HMC iteration

1. Draw  $\phi \sim N(0, M)$ .
2. Run  $L$  leapfrog steps from  $(\theta^{t-1}, \phi)$  to get  $(\theta^*, \phi^*)$ .
3. Accept  $\theta^*$  with probability

$$\min\left(1, \frac{p(\theta^* | y) p(\phi^*)}{p(\theta^{t-1} | y) p(\phi)}\right) = \min(1, \exp\{H(\theta^{t-1}, \phi) - H(\theta^*, \phi^*)\}),$$

otherwise keep  $\theta^{t-1}$ . (Strictly,  $\phi^*$  is negated before the acceptance step to ensure reversibility; this has no effect when  $p(\phi)$  is symmetric.)

The acceptance step corrects for discretization error; with a good  $\epsilon$ ,  $H$  is nearly conserved and acceptance is high.

### 4.4 Tuning

- **Mass matrix  $M$ :** ideally  $M^{-1} \approx$  posterior covariance (so  $M$  is the posterior precision), estimated during warm-up.
- **Step size and trajectory length:** set  $\epsilon L \approx 1$  on the scale of the posterior; [1] suggests targeting an acceptance rate of about 65%.
- Randomizing  $\epsilon$  and/or  $L$  avoids periodic behaviour.
- The No-U-Turn Sampler (NUTS) chooses  $L$  automatically by stopping a trajectory when it begins to double back; it is Stan's default.

*Remark.* HMC requires  $\nabla \log p(\theta | y)$  and a continuous parameter space. Discrete parameters must be marginalized out. [2] discusses Langevin/hybrid algorithms, which are the closest relatives treated there.

## References

- [1] Gelman, A., Carlin, J. B., Stern, H. S., Dunson, D. B., Vehtari, A., and Rubin, D. B. (2013). *Bayesian Data Analysis*, 3rd edition. Chapman & Hall/CRC, Boca Raton, FL. Chapter 12 (pp. 285–306).
- [2] Robert, C. P. and Casella, G. (2004). *Monte Carlo Statistical Methods*, 2nd edition. Springer, New York.